

Usman Rasheed

Software Engineer

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Portfolio <https://manisportfolio1.vercel.app/>

Objective

Passionate Software Engineer specializing in cross-platform mobile application development, fullstack solutions, and machine learning integration. Proven experience building scalable web platforms and AI powered mobile apps, including predictive computer vision systems for smart agriculture. Adept at bridging the gap between complex AI models and user-friendly digital interface

Experience

Mobile App Developer | Bit Built

2023-25

- Developed scalable, cross-platform mobile applications using React Native, focusing on high-performance interfaces and delivering real-world impact for end-users.
- Engineered AI-powered mobile applications, notably a Plant Disease Detection system for smart farming, seamlessly integrating machine learning inference models directly into the mobile ecosystem.
- Designed and managed secure backend infrastructure and inventory data using PostgreSQL and FastAPI, implementing strict Row Level Security (RLS) policies for robust data management.

Full Stack Developer | Brain X ltd abbt

2022-23

- Built scalable, efficient, and user-friendly web systems utilizing both frontend and backend technologies.
- Collaborated directly with clients to gather requirements, seamlessly integrating third-party APIs and delivering complete end-to-end software solutions.

- Developed and trained custom machine learning models tailored to specific client requirements, deploying practical and intelligent AI solutions for real-world business challenges.

Education

Pak-Austria Fachhochschule University

Bachelor of Science (B.S.) in Software Engineering

Higher Education (H.S.S.C) in Computer Science

Major: Mathematics | Minor: Computer Science

Secondary School Education (S.S.C) in Computer Science

Major: Mathematics | Minor: Computer Science

Skills & abilities

- **Languages:** Python, SQL.
- **Mobile & Frontend:** React Native.
- **Backend & Database:** FastAPI, PostgreSQL, MySQL, Database Management, Row Level Security (RLS).
- **AI & Machine Learning:** Machine Learning model training & deployment, Hugging Face, Plant Village Dataset, AI Prompt Engineering.
- **Core Competencies:** Full Stack Development, Creative Problem Solving, Critical Thinking.

PROJECTS & RESEARCH

- **AI Crop Disease Detection for Farmers**
Developed a React Native mobile application for comprehensive agricultural health monitoring. Trained image recognition models leveraging the Plant Village dataset to accurately identify plant diseases, and integrated Google Weather APIs to provide crucial environmental context for farmers.
- **Research Publication: "Model of Knowledge Sharing in Distributed Teams"**

Authored foundational research analyzing collaborative workflows in modern software engineering environments.

- **Solar Plate Defect Detection**

Utilized machine learning computer vision techniques to automatically identify physical defects in solar panels, improving maintenance efficiency and energy output.

- **Disaster Detection AI**

Designed and implemented an AI-based early warning system to detect and predict regional environmental disasters.